Tanta University

Faculty of Computers and Information

Question Bank 1

Deep Learning I: Foundations

CS433 – Selected Topics in Artificial Intelligence

Complete Solved Bank
30 Multiple-Choice Questions + 10 True/False Questions
Original English questions and options are preserved, followed by the correct answer, an academic explanation in English, and a simplified explanation in Arabic.

Prepared for study and revision

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1 Part A – Multiple Choice

Choose the one best answer.

A1. In the term “deep learning”, what does the word deep refer to?

(a) The large size of the training data set

(b) The long time needed to train the network

(c) The number of layers between the input and the output

(d) The very large values of the weights

Correct Answer: (c) The number of layers between the input and the output.

Explanation (English): The term *deep* describes the depth of the neural network: the number of layers between its input and output. It does not refer to the size of the data set, training time, or magnitude of the weights.

الشرح بالعربي: كلمة Deep معناها إن الشبكة العصبية فيها طبقات متعددة بين طبقة الإدخال وطبقة الإخراج. المقصود عمق الشبكة بعدد الطبقات، مش حجم البيانات أو وقت التدريب.

Reference: CS433-W04-Lecture-Notes, Section 4.1 – What is deep learning?

A2. What is meant by representation learning?

(a) The network learns useful features from raw data by itself, layer by layer

(b) Human experts design features, and a simple algorithm uses them

(c) The network stores the training examples so it can repeat them later

(d) The data is converted into a table before learning starts

Correct Answer: (a) The network learns useful features from raw data by itself, layer by layer.

Explanation (English): Representation learning means that the network automatically discovers useful features from raw input data. Different layers learn increasingly useful representations instead of relying entirely on manually designed features.

الشرح بالعربي: يعني الشبكة بتتعلم بنفسها تستخرج الخصائص المهمة من البيانات الخام، طبقة ورا طبقة، بدل ما الإنسان يحدد لها الخصائص يدويًا.

Reference: CS433-W04-Lecture-Notes, Sections 4.1–4.2.

A3. Deep networks took off around 2011 for three reasons. Which of the following is not one of them?

(a) Huge labelled data sets from the internet, such as ImageNet

(b) Fast matrix multiplication on graphics cards (GPUs)

(c) Better methods such as ReLU, dropout and batch normalisation

(d) The invention of the backpropagation algorithm in 2011

Correct Answer: (d) The invention of the backpropagation algorithm in 2011.

Explanation (English): Backpropagation was not invented in 2011. The rise of deep learning was supported by large data sets, GPU computing, and improved training methods and architectures.

الشرح بالعربي: الاختيار (d) هو اللي مش من الأسباب، لأن Backpropagation أقدم بكتير من 2011. الانتشار القوي للتعلم العميق حصل بسبب توافر بيانات ضخمة، ومعالجات GPU السريعة، وتحسن طرق التدريب زي ReLU وDropout وBatch Normalisation.

Reference: CS433-W04-Lecture-Notes, Section 4.1 and historical overview.

A4. What event is described in the notes as the turning point for deep learning?

(a) The proposal of the first artificial neuron in 1943

(b) The 2012 ImageNet competition, won by a deep convolutional network

(c) Minsky and Papert's book on single-layer networks in 1969

(d) The victory of AlphaGo at the game of Go

Correct Answer: (b) The 2012 ImageNet competition, won by a deep convolutional network.

Explanation (English): The 2012 ImageNet result demonstrated the strong performance of deep convolutional networks in image classification and is described in the notes as a major turning point.

الشرح بالعربي: المقصود هو مسابقة ImageNet سنة 2012، لما شبكة Convolutional عميقة حققت نتيجة قوية جدًا في تصنيف الصور، وده كان نقطة تحول مهمة في انتشار Deep Learning.

Reference: CS433-W04-Lecture-Notes, Section 4.1.

A5. A unit has inputs $x = (2, -1)$, weights $w = (0.5, 1.0)$, bias $b = 0.5$ and ReLU activation. What is its output?

(a) 0.5

(b) 0

(c) 1.5

(d) -0.5

Correct Answer: (a) 0.5.

Explanation (English): First calculate the weighted sum: $z = (2)(0.5) + (-1)(1.0) + 0.5 = 1 - 1 + 0.5 = 0.5$. ReLU is $\max(0, z)$, so the output is $\max(0, 0.5) = 0.5$.

الشرح بالعربي: بنحسب الأول المجموع الموزون: اتنين في 0.5 ناقص واحد في 1 زائد 0.5، الناتج 0.5. ودالة ReLU بترجع أكبر قيمة بين صفر وz، فالناتج النهائي 0.5.

Reference: CS433-W04-Lecture-Notes, Section 4.2 – The artificial neuron and ReLU.

A6. What is the role of the bias b in a unit?

(a) It decides how important each input is

(b) It sets the size of each gradient-descent step

(c) It shifts the point at which the unit "switches on"

(d) It switches the unit off at random during training

Correct Answer: (c) It shifts the point at which the unit "switches on".

Explanation (English): The bias is added to the weighted sum, $z = w \cdot x + b$. It shifts the activation threshold. The weights determine how strongly individual inputs contribute.

الشرح بالعربي: الـ Bias قيمة بتتضاف للمجموع الموزون، وبتحرّك نقطة التفعيل أو النقطة اللي عندها الـ neuron يبدأ يطلع قيمة موجبة. أما أهمية كل input فبتتحدد من الـ Weights.

Reference: CS433-W04-Lecture-Notes, Section 4.2.

A7. What is the output range of the tanh activation function?

(a) 0 to 1

(b) 0 to ∞

(c) 0 to 1, summing to 1

(d) -1 to 1

Correct Answer: (d) -1 to 1.

Explanation (English): The hyperbolic tangent function, $\tanh(z)$, produces values in the interval $[-1, 1]$.

الشرح بالعربي: دالة tanh مخرجاتها بتكون من سالب واحد لحد موجب واحد، وده الفرق الأساسي بينها وبين Sigmoid اللي مخرجاتها من صفر لواحد.

Reference: CS433-W04-Lecture-Notes, Section 4.3 – Activation functions.

A8. Which activation function is used at the output of a multi-class classifier?

(a) ReLU

(b) Softmax

(c) tanh

(d) Sigmoid

Correct Answer: (b) Softmax.

Explanation (English): Softmax converts the class logits into probabilities between 0 and 1 whose sum is 1, making it suitable for multi-class classification.

الشرح بالعربي: Softmax بتحول درجات الفئات (Logits) لاحتمالات، كل احتمال بين صفر وواحد، ومجموع الاحتمالات كلها بيساوي واحد؛ عشان كده بنستخدمها في التصنيف متعدد الفئات.

Reference: CS433-W04-Lecture-Notes, Section 4.3.

A9. Why is ReLU the default activation for hidden layers today?

(a) Its output always lies between 0 and 1

(b) It gives probabilities that sum to 1

(c) It is cheap to compute and its slope is exactly 1 for every positive input

(d) Its slope is at most 0.25, which keeps the gradients small

Correct Answer: (c) It is cheap to compute and its slope is exactly 1 for every positive input.

Explanation (English): ReLU is computationally inexpensive. For positive inputs its derivative is 1, which helps gradients pass through deep networks without the systematic shrinking associated with sigmoid activations.

الشرح بالعربي: ReLU سهلة وسريعة في الحساب، وميلها بيساوي واحد لما المدخل موجب. ده بيساعد الـ Gradients إنها تعدّي خلال الطبقات بدل ما تصغر جدًا زي ما ممكن يحصل مع Sigmoid.

Reference: CS433-W04-Lecture-Notes, Section 4.3.

A10. What happens if we remove the activation functions and stack many linear layers?

(a) The whole network is still just one linear function

(b) The network can learn any curved decision boundary

(c) The network trains faster and generalises better

(d) The network can learn XOR but not other problems

Correct Answer: (a) The whole network is still just one linear function.

Explanation (English): A composition of linear functions is itself a linear function. Therefore, stacking linear layers without non-linear activations does not increase the network's ability to represent non-linear decision boundaries.

الشرح بالعربي: تركيب دوال خطية فوق بعض بيفضل في النهاية دالة خطية واحدة. يعني حتى لو زودنا عدد الطبقات، من غير Activation غير خطية مش هنقدر نمثل حدود قرار معقدة أو غير خطية.

Reference: CS433-W04-Lecture-Notes, Section 4.3 – Why we need non-linearity.

A11. Which problem did Minsky and Papert use in 1969 to show the limits of single-layer networks?

(a) Handwritten digit recognition

(b) Face recognition

(c) House-price prediction

(d) XOR

Correct Answer: (d) XOR.

Explanation (English): The XOR problem is not linearly separable. A single-layer perceptron cannot represent its decision boundary, illustrating the limitations of single-layer networks.

الشرح بالعربي: مشكلة XOR مش ممكن نفصل نتايجها بخط مستقيم واحد؛ يعني مش Linearly Separable. وده وضّح محدودية الشبكات ذات الطبقة الواحدة.

Reference: CS433-W04-Lecture-Notes, Section 4.3.

A12. In a feedforward network, which layer does no computation?

(a) The first hidden layer

(b) The input layer

(c) The output layer

(d) The last hidden layer

Correct Answer: (b) The input layer.

Explanation (English): The input layer receives and passes the input values to the next layer. The weighted sums and activation computations are performed by subsequent layers.

الشرح بالعربي: طبقة الإدخال وظيفتها تستقبل البيانات وتعدّيها للطبقة اللي بعدها؛ الحسابات الفعلية زي ضرب الأوزان وجمع الـ Bias وتطبيق Activation بتحصل في الطبقات التالية.

Reference: CS433-W04-Lecture-Notes, Section 4.4 – Feedforward networks.

A13. A network takes a 28 × 28 image (784 pixels), has one hidden layer of 128 units and outputs 10 classes. How many parameters does it have?

(a) 100,480

(b) 1,290

(c) 101,770

(d) 101,632

Correct Answer: (c) 101,770.

Explanation (English): For a fully connected layer, parameters equal weights plus biases. Input-to-hidden: $784 \times 128 + 128 = 100,480$. Hidden-to-output: $128 \times 10 + 10 = 1,290$. Total: $100,480 + 1,290 = 101,770$.

الشرح بالعربي: كل طبقة Fully Connected عدد بارامتراتها = عدد الأوزان + عدد الـ Biases. من الإدخال للمخفية: $784 \times 128 + 128 = 100480$. ومن المخفية للإخراج: $128 \times 10 + 10 = 1290$. الإجمالي = 101770.

Reference: CS433-W04-Lecture-Notes, Section 4.4 – Counting parameters.

A14. How many parameters does a fully connected network with layers 100 → 50 → 10 have?

(a) 5,560

(b) 5,500

(c) 5,050

(d) 5,060

Correct Answer: (a) 5,560.

Explanation (English): For the first layer: $100 \times 50 + 50 = 5,050$. For the second layer: $50 \times 10 + 10 = 510$. Total parameters: $5,050 + 510 = 5,560$.

الشرح بالعربي: الطبقة الأولى فيها 100×50 وزن و50 Bias، يعني 5050. الطبقة التانية فيها 50×10 وزن و10 Bias، يعني 510. المجموع 5560 بارامتر.

Reference: CS433-W04-Lecture-Notes, Section 4.4.

A15. What does the universal approximation theorem say?

(a) Any network with ReLU units will reach zero training loss

(b) A network with just one hidden layer, given enough units, can approximate any continuous function as closely as we like

(c) Deep networks always need fewer training examples than shallow ones

(d) Gradient descent always finds the global minimum of the loss

Correct Answer: (b) A network with just one hidden layer, given enough units, can approximate any continuous function as closely as we like.

Explanation (English): The universal approximation theorem states that a sufficiently wide network with one hidden layer can approximate continuous functions to arbitrary accuracy under the theorem's conditions. It does not guarantee successful training or a global minimum.

الشرح بالعربي: النظرية بتقول إن شبكة فيها Layer Hidden واحدة، لو عدد الوحدات فيها كافي، تقدر تقرّب أي دالة مستمرة بدرجة دقة كبيرة. لكنها مش معناها إن التدريب مضمون يوصل لأقل Loss أو إن Descent Gradient يلاقي الحل الأمثل دائمًا.

Reference: CS433-W04-Lecture-Notes, Section 4.4.

A16. In the forward-pass example, the input is $x = (1, 2)$ and hidden unit h_2 has weights $(-1, 0.5)$, bias 0.5 and ReLU activation. What is the value of h_2?

(a) 0

(b) 1.0

(c) 1.5

(d) 0.5

Correct Answer: (d) 0.5.

Explanation (English): Compute $z = (1)(-1) + (2)(0.5) + 0.5 = -1 + 1 + 0.5 = 0.5$. Applying ReLU gives $h_2 = \max(0, 0.5) = 0.5$.

الشرح بالعربي: نحسب: واحد في سالب واحد، زائد اتنين في 5.0، زائد 5.0. الناتج 5.0. وبما إن ReLU بترجع الأكبر بين الصفر والقيمة، يبقى $h_2 = 0.5$.

Reference: CS433-W04-Lecture-Notes, Section 4.4 – Forward-pass example.

A17. In the same example, $h_1 = 1.0$ and $h_2 = 0.5$. The sigmoid output unit has weights $(1, -2)$ and bias 0.5. What is $\hat{y}$?

(a) 0.50

(b) 0.73

(c) 0.62

(d) 0.38

Correct Answer: (c) 0.62.

Explanation (English): The output logit is $z = (1)(1.0) + (-2)(0.5) + 0.5 = 1 - 1 + 0.5 = 0.5$. The sigmoid gives $\sigma(0.5) = 1/(1 + e^{-0.5}) \approx 0.622$, so the closest option is 0.62.

الشرح بالعربي: قيمة الـ logit = واحد في 1 ناقص اتنين في 5.0 زائد 5.0، يعني 5.0. نطبق Sigmoid على 5.0 فنحصل على حوالي 0.622، وأقرب اختيار هو 0.62.

Reference: CS433-W04-Lecture-Notes, Section 4.4.

A18. For a binary (yes/no) classification task, which output layer and loss should be used?

(a) One sigmoid unit with binary cross-entropy

(b) One linear unit with mean squared error

(c) Softmax over K units with categorical cross-entropy

(d) One ReLU unit with mean squared error

Correct Answer: (a) One sigmoid unit with binary cross-entropy.

Explanation (English): A binary classifier uses one sigmoid output to represent the probability of the positive class and binary cross-entropy to measure the prediction error.

الشرح بالعربي: في تصنيف Yes/No بنستخدم Sigmoid واحدة عشان تطلع احتمال الفئة الموجبة، ومعاها Cross-Entropy Binary لحساب الخطأ بين التوقع والقيمة الصحيحة.

Reference: CS433-W04-Lecture-Notes, Section 4.5 – Output layers and losses.

A19. True prices are (3, 5, 2) and the predictions are (2.5, 5.5, 2). What is the mean squared error?

(a) 0.5

(b) 0.333

(c) 0.25

(d) 0.167

Correct Answer: (d) 0.167.

Explanation (English): The squared errors are $(3-2.5)^2 = 0.25$, $(5-5.5)^2 = 0.25$, and $(2-2)^2 = 0$. Thus, $MSE = (0.25+0.25+0)/3 = 0.5/3 \approx 0.167$.

الشرح بالعربي: بنربع الفرق بين كل قيمة حقيقية والتوقع: 25.0 و25.0 وصفر. نجمعهم = 5.0، ونقسم على عدد القيم 3، فيكون الـ MSE حوالي 0.167.

Reference: CS433-W04-Lecture-Notes, Section 4.5 – Mean squared error.

A20. A 3-class network outputs logits $z = (2.0, 1.0, 0.1)$. Given $e^{2.0} = 7.389$, $e^{1.0} = 2.718$ and $e^{0.1} = 1.105$, what is the softmax probability of class 1?

(a) 0.242

(b) 0.659

(c) 0.667

(d) 0.099

Correct Answer: (b) 0.659.

Explanation (English): For class 1, softmax is $e^{2.0}/(e^{2.0}+e^{1.0}+e^{0.1})$. Substituting the values gives $7.389/(7.389+2.718+1.105) = 7.389/11.212 \approx 0.659$.

الشرح بالعربي: احتمال Class 1 = قيمة الأس الخاصة بيها مقسومة على مجموع قيم الأس لكل الفئات. يعني 7.389 على مجموع $7.389+2.718+1.105$، والناتج تقريبًا 0.659.

Reference: CS433-W04-Lecture-Notes, Section 4.5 – Softmax and categorical cross-entropy.

A21. A classifier gives the correct class a probability of 0.1. What is the cross-entropy loss?

(a) 0.105

(b) 0.9

(c) 2.303

(d) 0.1

Correct Answer: (c) 2.303.

Explanation (English): For a correct class probability p, the cross-entropy loss for one example is $-\log(p)$. Therefore, $-\log(0.1) \approx 2.303$.

الشرح بالعربي: قانون الـ Cross-Entropy للحالة دي هو سالب لوغاريتم احتمال الفئة الصحيحة. بما إن الاحتمال 1.0، يبقى $-\ln(0.1) = 2.303$. الاحتمال القليل للفئة الصحيحة معناه Loss كبير.

Reference: CS433-W04-Lecture-Notes, Section 4.5.

A22. Which rule is the gradient-descent update for a weight w with learning rate η?

(a) $w \leftarrow w - \eta$
$, \partial L/\partial w$

(b) $w \leftarrow w + \eta$
$, \partial L/\partial w$

(c) $w \leftarrow \eta$
$, \partial L/\partial w$

(d) $w \leftarrow w - \partial L/\partial \eta$

Correct Answer: (a) $w \leftarrow w - \eta$
$, \partial L/\partial w$.

Explanation (English): Gradient descent updates a parameter by subtracting the learning rate multiplied by the loss gradient. The negative sign moves the parameter in the direction that locally decreases the loss.

الشرح بالعربي: بنحدّث الوزن عن طريق طرح الـ Rate Learning مضروبة في مشتقة الـ Loss بالنسبة للوزن. علامة السالب مهمة لأنها بتحركنا عكس اتجاه زيادة الخطأ، وبالتالي نقلل الـ Loss.

Reference: CS433-W04-Lecture-Notes, Section 4.6 – Gradient descent.

A23. What is an epoch?

(a) One update of the weights

(b) A small random group of examples, such as 32 or 128

(c) The size of each gradient-descent step

(d) One pass through the whole training set

Correct Answer: (d) One pass through the whole training set.

Explanation (English): An epoch is one complete pass through all training examples. It usually consists of multiple mini-batches and parameter updates.

الشرح بالعربي: الـ Epoch معناها إن الموديل مرّ على كل بيانات التدريب مرة كاملة. وغالبًا الـ Epoch الواحدة بتتكون من Mini-batches كتير، وكل واحدة ممكن يحصل بعدها تحديث للأوزان.

Reference: CS433-W04-Lecture-Notes, Section 4.6.

A24. Backpropagation is best described as:

(a) Trying all combinations of weights and keeping the best

(b) The chain rule of calculus, applied systematically from the output back to the input

(c) Randomly switching off hidden units during training

(d) Adding a penalty for large weights to the loss

Correct Answer: (b) The chain rule of calculus, applied systematically from the output back to the input.

Explanation (English): Backpropagation applies the chain rule through the network's computation graph, propagating derivatives from the output toward earlier layers to calculate gradients for the parameters.

الشرح بالعربي: Backpropagation بتستخدم Rule Chain عشان تحسب تأثير كل وزن على الخطأ. بنبدأ من الـ Output ونرجع للخلف طبقة طبقة، ونحسب الـ Gradients اللي هنستخدمها في تحديث الأوزان.

Reference: CS433-W04-Lecture-Notes, Section 4.7 – Backpropagation.

A25. In the training step on one neuron ($x = 2$, $y = 1$, $w = 0.5$, $b = 0$, sigmoid, cross-entropy), the forward pass gives $a = 0.731$. Using $\partial L/\partial z = a - y$, what is $\partial L/\partial w$?

(a) -0.269

(b) 0.538

(c) -0.538

(d) 1.462

Correct Answer: (c) -0.538.

Explanation (English): First, $\partial L/\partial z = a - y = 0.731 - 1 = -0.269$. Since $z = wx + b$, the chain rule gives $\partial L/\partial w = (\partial L/\partial z)(\partial z/\partial w) = (-0.269)(x) = (-0.269)(2) = -0.538$.

الشرح بالعربي: أولًا مشتقة الخطأ بالنسبة لـ z تساوي $0.731 - 1 = -0.269$. وبما إن $z = wx + b$، فمشتقة z بالنسبة للوزن هي $x = 2$. إذن مشتقة الـ Loss بالنسبة للوزن = $-0.269 \times 2 = -0.538$.

Reference: CS433-W04-Lecture-Notes, Section 4.7 – One training step.

A26. Continuing A25 with learning rate $\eta = 0.5$, what is the new value of w?

(a) 0.769

(b) 0.231

(c) 1.038

(d) 0.634

Correct Answer: (a) 0.769.

Explanation (English): Use $w_{new} = w_{old} - \eta(\partial L/\partial w)$. Therefore, $w_{new} = 0.5 - 0.5(-0.538) = 0.5 + 0.269 = 0.769$.

الشرح بالعربي: قانون التحديث: الوزن الجديد = الوزن القديم ناقص Rate Learning في Gradient. بالتعويض: $0.5 - 0.5(-0.538) = 0.769$. لأن الـ Gradient سالب، الطرح منه خلّى الوزن يزيد.

Reference: CS433-W04-Lecture-Notes, Section 4.7.

A27. The sigmoid's slope is at most 0.25. In a 10-layer sigmoid network, how small can the gradient reaching the first layer be?

(a) About 2.5

(b) About 0.025

(c) About 0.000001

(d) Exactly 0

Correct Answer: (c) About 0.000001.

Explanation (English): In the worst shrinking case, the gradient is multiplied by a factor no larger than 0.25 at each of 10 layers: $0.25^{10} \approx 9.54 \times 10^{-7}$, which is about 0.000001. This illustrates vanishing gradients.

الشرح بالعربي: كل طبقة ممكن تضرب الـ Gradient في قيمة أقصاها 0.25. خلال 10 طبقات: 0.25^{10}، والناتج تقريبًا 0.000001. كده الإشارة اللي بتوصل للطبقات الأولى بتكون صغيرة جدًا، ودي مشكلة Vanishing Gradient.

Reference: CS433-W04-Lecture-Notes, Section 4.7 – The vanishing gradient problem.

A28. How is the problem of exploding gradients controlled?

(a) Using sigmoid activations

(b) Setting all initial weights equal

(c) Removing the bias terms

(d) Gradient clipping

Correct Answer: (d) Gradient clipping.

Explanation (English): Gradient clipping limits the magnitude of gradients when they become excessively large, helping prevent unstable updates caused by exploding gradients.

الشرح بالعربي: Clipping Gradient معناها بنحط حد أقصى لحجم الـ Gradient. لو المشتقة طلعت كبيرة جدًا، بنقللها للحد المحدد عشان تحديث الأوزان مايبقاش عنيف وغير مستقر.

Reference: CS433-W04-Lecture-Notes, Section 4.7.

A29. What is special about the Adam optimiser?

(a) It uses the same fixed step size for every weight

(b) It adapts the step size for each weight separately, using running averages of the gradients and their squares

(c) It computes the loss over the whole training set before every step

(d) It switches off hidden units with probability p

Correct Answer: (b) It adapts the step size for each weight separately, using running averages of the gradients and their squares.

Explanation (English): Adam combines running estimates of the first moment (the gradient) and second moment (the squared gradient) to adapt parameter updates. This gives each parameter an adaptive effective step size.

الشرح بالعربي: Adam بيحتفظ بمتوسطات متحركة لـ Gradients ومربعاتها، وبيستخدمهم عشان يكيّف حجم خطوة التحديث لكل وزن. يعني مش بيستخدم نفس خطوة التحديث الثابتة لكل الأوزان.

Reference: CS433-W04-Lecture-Notes, Section 4.8 – Practical training and optimisers.

A30. Which statement about dropout is correct?

(a) During training each hidden unit is switched off with probability p, and at test time all units are used

(b) Units are switched off at test time to make predictions faster

(c) It removes the weights with the largest values after training

(d) It stops training when the validation loss starts to rise

Correct Answer: (a) During training each hidden unit is switched off with probability p, and at test time all units are used.

Explanation (English): Dropout randomly disables units during training to reduce co-adaptation and overfitting. At test time, the full network is used, with the appropriate scaling handled by the dropout implementation.

الشرح بالعربي: أثناء التدريب بنعطّل بعض الـ Units عشوائيًا باحتمال p، وده بيقلل اعتماد الوحدات على بعضها وبيساعد في تقليل Overfitting. وقت الاختبار بنستخدم الشبكة كاملة.

Reference: CS433-W04-Lecture-Notes, Section 4.9 – Regularisation and dropout.

2 Part B – True or False

State whether each statement is true or false.

B1. The first artificial neuron was proposed by McCulloch and Pitts in 1943.

Correct Answer: True.

Explanation (English): McCulloch and Pitts proposed an early mathematical model of an artificial neuron in 1943.

الشرح بالعربي: العبارة صح. McCulloch وPitts قدموا نموذجًا رياضيًا مبكرًا للخلية العصبية الاصطناعية سنة 1943.

Reference: CS433-W04-Lecture-Notes, historical introduction to neural networks.

B2. In a feedforward network, information can flow in loops from the output back to the input during the forward pass.

Correct Answer: False.

Correct statement: In a feedforward network, information flows from the input toward the output without loops during the forward pass.

Explanation (English): A feedforward network passes information in one direction, from input through hidden layers to output. Feedback loops are not part of the feed-forward computation.

الشرح بالعربي: العبارة غلط، لأن كلمة Feedforward معناها إن البيانات بتتحرك في اتجاه واحد: من Input إلى Hidden Layers، ثم Output، من غير حلقات راجعة أثناء الـ Forward Pass.

Reference: CS433-W04-Lecture-Notes, Section 4.4.

B3. The cross-entropy loss is 0 when the network gives the correct class a probability of 0.

Correct Answer: False.

Correct statement: The cross-entropy loss becomes very large (tends to infinity) when the correct class is assigned a probability of 0; it is 0 when that probability is 1.

Explanation (English): For the correct class, cross-entropy is $-\log(p)$. At $p = 0$, the loss tends to infinity; at $p = 1$, it equals zero.

الشرح بالعربي: العبارة غلط. القانون هو سالب لوغاريتم احتمال الفئة الصحيحة. لو الاحتمال صفر فال Loss بيكون كبير جدًا ويميل إلى ما لا نهاية، أما لو الاحتمال واحد فال Loss يساوي صفر.

Reference: CS433-W04-Lecture-Notes, Section 4.5.

B4. Mini-batch gradients are noisy, but the noise can help the network escape poor regions of the loss landscape.

Correct Answer: True.

Explanation (English): A mini-batch estimates the gradient using only a subset of the training data, so its gradient can be noisy. This noise can sometimes help optimisation move out of poor regions or shallow local structures.

الشرح بالعربي: العبارة صح. لأن الـ Mini-batch بيستخدم جزء من البيانات مش كلها، فال Gradient بيكون فيه شوية Noise. أحيانًا الضوضاء دي بتساعد الموديل يخرج من مناطق سيئة في سطح الـ Loss.

Reference: CS433-W04-Lecture-Notes, Section 4.6 – Mini-batch SGD.

B5. The backward pass of backpropagation costs far more than the forward pass, which is why it cannot be used for networks with billions of weights.

Correct Answer: False.

Correct statement: Backpropagation is computationally practical because the backward pass has a cost of the same general order as the forward pass; it is routinely used to train large neural networks.

Explanation (English): Backpropagation efficiently computes gradients by reusing intermediate values from the forward computation. The backward pass is not prohibitively more expensive than the forward pass, so large networks can be trained with it.

الشرح بالعربي: العبارة غلط. الـ Backward Pass مش أغلى بدرجة تمنع استخدامه؛ تكلفته من نفس الرتبة العامة لتكلفة الـ Forward Pass. وده السبب إننا بنقدر ندرّب شبكات كبيرة باستخدام Backpropagation.

Reference: CS433-W04-Lecture-Notes, Section 4.7.

B6. If all the weights in a layer started with the same value, all the units in that layer would compute the same thing and learn the same thing.

Correct Answer: True.

Explanation (English): Identical initial weights create symmetry: units receiving the same inputs produce the same outputs and gradients, so they remain identical instead of learning different features. This is why suitable random initialisation is important.

الشرح بالعربي: العبارة صح. لو كل الـ Weights بدأت بنفس القيمة، الوحدات المتشابهة هتطلع نفس النتائج وتاخد نفس الـ Gradients، وبالتالي هتتعلم نفس الحاجة. عشان كده بنستخدم Initialisation مناسب يكسر التماثل ده.

Reference: CS433-W04-Lecture-Notes, Section 4.8 – Initialisation.

B7. Batch normalisation re-scales the values inside the network for each mini-batch to mean 0 and variance 1, followed by a learned scale and shift.

Correct Answer: True.

Explanation (English): Batch normalisation normalises activations using the mini-batch mean and variance, then applies learned scale and shift parameters.

الشرح بالعربي: العبارة صح. Normalisation Batch بتطبّع القيم باستخدام متوسط وتباين الـ Mini-batch، وبعدها بتطبق معامل Scale وقيمة Shfit بيتعلمهم الموديل.

Reference: CS433-W04-Lecture-Notes, Section 4.8.

B8. The test set should be checked after every epoch to decide when to stop training.

Correct Answer: False.

Correct statement: The validation set should be monitored during training to decide when to stop; the test set should be reserved for final evaluation.

Explanation (English): Repeatedly using the test set to make training decisions leaks information from the test data into model development. Early stopping should monitor validation performance, while the test set is kept for an unbiased final evaluation.

الشرح بالعربي: العبارة غلط. اللي تتابعه بعد كل Epoch عشان نقرر نوقف التدريب هو Validation Set. إنما Test Set بنسيبه للتقييم النهائي فقط، عشان ما نستخدمش معلوماته في قرارات التدريب.

Reference: CS433-W04-Lecture-Notes, Section 4.8 – Training, validation and test split; early stopping.

B9. Weight decay (L2 regularisation) adds the penalty $\lambda \sum_i w_i^2$ to the loss to keep the weights small.

Correct Answer: True.

Explanation (English): L2 regularisation adds a penalty proportional to the sum of squared weights. This discourages excessively large weights and can reduce overfitting.

الشرح بالعربي: العبارة صح. Decay Weight أو Regularisation L2 بتضيف للـ Loss عقوبة تساوي لامبدا في مجموع مربعات الأوزان، وده بيشجع الأوزان إنها تفضل صغيرة ويقلل الـ Overfitting.

Reference: CS433-W04-Lecture-Notes, Section 4.9 – Regularisation.

B10. In early stopping, training is stopped when the training loss starts to rise.

Correct Answer: False.

Correct statement: In early stopping, training is stopped when the validation loss stops improving (or begins to rise), according to the chosen patience rule.

Explanation (English): Training loss may continue to decrease while the model overfits. Early stopping therefore monitors validation loss and stops when validation performance no longer improves.

الشرح بالعربي: العبارة غلط. مش بنوقف التدريب لمجرد إن Loss Training بدأ يزيد، غالبًا هو بيستمر في النزول حتى مع حدوث Overfitting. بنراقب Validation Loss، ولما يبطل يتحسن أو يبدأ يزيد بنوقف التدريب حسب قاعدة الـ Patience.

Reference: CS433-W04-Lecture-Notes, Section 4.9 – Early stopping.

3 Quick Answer Key

Part A – Multiple Choice

Question	Answer	Question	Answer	Question	Answer	Question	Answer
A1	C	A2	A	A3	D	A4	B
A5	A	A6	C	A7	D	A8	B
A9	C	A10	A	A11	D	A12	B
A13	C	A14	A	A15	B	A16	D
A17	C	A18	A	A19	D	A20	B
A21	C	A22	A	A23	D	A24	B
A25	C	A26	A	A27	C	A28	D
A29	B	A30	A				

Part B – True or False

Question	Answer	Question	Answer	Question	Answer	Question	Answer
B1	True	B2	False	B3	False	B4	True
B5	False	B6	True	B7	True	B8	False
B9	True	B10	False				

End of Question Bank 1 – Deep Learning I: Foundations